Homework 5- Solutions

Problem 1. Sarah works as a sales representative selling house insurance over the phone. Only 10% of all calls result in a successful sale. Assume that the outcome of each call is independent of the others.

(a) What is the probability that Sarah makes her first sale on an odd-numbered call?

(b) What is the probability that Sarah makes her second sale on an even-numbered call?

(a) X =number of calls till first sale
$X \sim Geom(p = 0.1)$

$$P(X=1)+P(X=3)+P(X=5)+\cdots=\sum_{k=1}^{\infty}P(X=2k-1)=\sum_{k=1}^{\infty}(1-p)^{2k-2}p$$

$$=p\sum_{k=1}^{\infty}((1-p)^2)^{k-1}=p\sum_{k=0}^{\infty}((1-p)^2)^{k}=\frac{p}{(1-(1-p)^2)}=\frac{p}{1-1-p^2+2p}=\frac{1}{2-p}$$

(b) Y =number of calls till second sale
$Y \sim \text{NBin}(r = 2, p = 0.1)$

$$P(Y=2)+P(Y=4)+\cdots=\sum_{k=1}^{\infty}P(Y=2k)=\sum_{k=1}^{\infty}\binom{2k-1}{1}(1-p)^{2k-2}p^2$$

$$=p^2\sum_{k=1}^{\infty}\binom{2k-1}{1}(1-p)^{2k-2}=p^2\sum_{k=1}^{\infty}(2k-1)(1-p)^{2k-2}=\frac{1+(1-p)^2}{(2-p)^2}$$

Problem 2. Let X and Y be two independent random variables, X has a Geometric distribution with the probability of "success" $p = 1/3$, Y has a Poisson distribution with mean 3.
Find $P(X = Y)$

$X \sim Geom(\frac{1}{3})$, $X = 1, 2, 3, \cdots$.
$X \sim Pois(\lambda = 3)$, $Y = 0, 1, 2, 3, \cdots$.
Since X and Y are independent.

$$P(X=x, Y=y) = P(x=x)P(Y=y)$$

$$P(X=Y) = \sum_{k=1}^{\infty} P(X=k)P(Y=k) = \sum_{k=1}^{\infty} (\frac{2}{3})^{k-1}(\frac{1}{3})e^{-3}\frac{3^k}{k!}$$
$$= e^{-3}\sum_{k=1}^{\infty}\frac{2^{k-1}}{3^{k-1}}\cdot\frac{1}{3}\cdot\frac{3^k}{k!} = e^{-3}\sum_{k=1}^{\infty}\frac{2^{k-1}}{k!}$$
$$= \frac{e^{-3}}{2}\sum_{k=1}^{\infty}\frac{2^{k-1}}{k!}\cdot 2 = \frac{e^{-3}}{2}\sum_{k=1}^{\infty}\frac{2^k}{k!}$$

We know Taylor series of $e^x = \sum_{k=0}^{\infty}\frac{x^k}{k!}$, hence, $e^2 = \sum_{k=0}^{\infty}\frac{2^k}{k!}$. Hence, $\sum_{k=1}^{\infty}\frac{2^k}{k!} + 1 = e^2$.

Therefore, $\sum_{k=1}^{\infty}\frac{2^k}{k!} = e^2 - 1$. Hence,

$$P(X=Y) = \frac{e^{-3}}{2}(e^2-1) = \frac{e^{-1}-e^{-3}}{2}$$

Problem 3. In this problem you may neglect the probability of twins.

(a) A family has 5 children. Find the probability that they have 2 boys and 3 girls.

(b) A family decides to have children until they have 3 girls. Find the probability that they have 2 boys.

(c) A family decides to have children until they have 3 girls. Let C be the total number of children in the family. Compute E[C] and V(C).

(a) In general for each child there are two possibilities; boy or girl. Hence, if the total number of children is n, then

X = Total number of Boys, $X \sim Bin(n, \frac{1}{2})$

$P(X=2) = \binom{5}{2}(\frac{1}{2})^5 = \frac{10}{2^5} = \frac{5}{16}$

(b) number of trials until they have three girls is follows a negative binomial distribution with $r = 3$, $p = \frac{1}{2}$.

The experiment stops when they have 3 girls, so having 2 boys in this experiment means in total they have 2 boys+ 3 girls = 5.

If X = number of trials till having 3 girls, $X \sim NBin(r = 3, p = \frac{1}{2})$.

Hence, the question is asking for

$P(X = 5) = \binom{4}{2}(\frac{1}{2})^5 = \frac{6}{32} = \frac{3}{16}$

(c) If X is the number of trials until they have 3 girls. $X \sim NBin(r = 3, p = \frac{1}{2})$. If you note in each trial a child is added to the number of children,

$$P(C = k) = P(X = k) \tag{1}$$

Hence, $E[C] = E[X]$, and $Var(X) = Var(C)$. Hence,

$E[C] = \frac{r}{p} = 6$, $Var(C) = r\frac{1-p}{p^2} = 3(\frac{1}{2})4 = 6$

Problem 4. Consider a random variable X with E[X] = 4 and Var[X] = 9.

(a) Calculate $P(|X - 1| < 3)$ if X follows a normal distribution. Write your final answer based on the CDF of standard normal.

(b) Find $P(1 < X^2 < 4)$. Write your final answer based on the CDF of standard normal.

$\mu = 4$, $\sigma^2 = 9$, $\sigma = 3$.

(a) $P(|X - 1| < 3)$

$|X - 1| < 3$, then $-3 < X - 1 < 3$. Simplifying it we get $-2 < X < 4$. Now we standardize X, we get $\frac{-2-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{4-\mu}{\sigma}$. We know

$Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$

$P(|X - 1| < 3) = P(-2 < Z < 0)$ and by 68-95-99.7 rule we know $P(-2 < Z < 2) = 0.95$, hence by symmetry around mean we have

$$P(|X-1|<3) = P(-2<Z<0) = \frac{0.95}{2} = 0.475$$

(b)

$$\begin{aligned}
P(1<X^2<4) &= P(1<X<2 \text{ or } -2<X<-1) \\
&= P(1<X<2) + P(-2<X<-1) \\
&= P(\frac{1-4}{3} < \frac{X-\mu}{\sigma} < \frac{2-4}{3}) + P(\frac{-2-4}{3} < \frac{X-\mu}{\sigma} < \frac{-1-4}{3}) \\
&= P(-1<Z<-\frac{2}{3}) + P(-2<Z<-\frac{5}{3}) \\
&= \Phi(-\frac{2}{3}) - \Phi(-1) + \Phi(-\frac{5}{3}) - \Phi(-2)
\end{aligned}$$

Problem 5. Between the hours of 2 and 4p.m. the average number of phone calls per minute coming into the switch-board of a company is 2.5. Find the probability that during one particular minute there will be
(a)0,
(b)1,
(c)2,
(d)3 or fewer,
(f) more than 3 phone calls.

Solutions. $\lambda = 2.5$

$$P(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}; k = 0, 1, 2, \cdots$$

(a) $P(X=0) = e^{-2.5}$

(b) $P(X=1) = 2.5\, e^{-2.5}$

(c) $P(X=2) = \frac{(2.5)^2}{2!}\, e^{-2.5}$

(d)$P(X \le 3) = e^{-2.5}(1 + 2.5 + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!})$

(f) $P(X>3) = 1 - p(X \le 3) = 1 - e^{-2.5}(1 + 2.5 + \frac{(2.5)^2}{2!} + \frac{(2.5)^3}{3!})$

Problem 6. The number of typos made by Sarah follows Poisson distribution with the rate of 0.4 typos per page. Assume that the numbers of typos on different pages are independent.

(a) Find the probability that there are exactly 4 typos in a 10-page lecture.

(b) Find the probability that there is at least one page with no typos in a 10-page lecture.

(c) Find the probability that there are exactly three pages with no typos in a 10-page lecture.

Problem 6 Solution. $\lambda = 0.4$ typoes/page

(a) Y = number of typos in a 10-page lecture. $Y = Pois(0.4 \times 10) = Pois(4)$.

$$P(Y = 4) = e^{-4}(\frac{4^4}{4!}) = e^{-4}(\frac{32}{3})$$

(b) number of pages with no typo in 10-page lecture. In this case each page is a trial, no typo= success.

$$P(\text{no typo in one page}) = P(X = 0) = e^{-0.4}$$

Hence, W= number of success in 10 trails, where success is no typo. Hence, $W \sim Bin(n = 10, e^{-0.4})$.

$$P(W \geq 1) = 1 - P(W = 0) = 1 - (1 - e^{-0.4})^{10}$$

(c)

$$P(W = 3) = \binom{10}{3}(e^{-0.4})^3(1 - e^{-0.4})^7$$